

The Intensive Margin of Small-Firm Protection: Bidding, Rents, and the Fiscal Cost of Public Procurement Set-Asides*

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Abstract

Using within-auction bid-distribution moments around a discrete policy cutoff, I establish that SME set-asides raise procurement prices primarily through the intensive margin of bidding, not through reduced entry. When Brazil’s March 2018 reinterpretation extended Sao Paulo’s SME-only regime to medical supplies—the only previously exempt product group—the winning bid moved ten percentage points closer to the buyer’s reference price, and the standard deviation of phase-2 submitted bids shrank monotonically in the number of firms, effects that persist when firm count is held fixed at two, three, or five-plus. A Gelbach decomposition assigns only six percent of the 12–13% price increase (7–10% real) to entry and winner composition; ninety-four percent reflects bidding behavior within auctions. Identification uses a single-cohort staggered DiD with 76 never-treated controls, robust to the Callaway-Sant’Anna, Sun-Abraham, and Goodman-Bacon estimators and to a formal spillover test. The fiscal cost for Group 65 alone is R\$50–85 million (US\$14–24 million) over 18 months, but 92% of procurement value concentrates in the top quartile: a value-threshold exemption recovers 90% of the cost while preserving the SME preference on 75% of items. The distortion is 62% larger in CMED price-regulated pharmaceuticals, exposing a perverse interaction between two Brazilian regulatory regimes.

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An additional bidder is worth more to the auctioneer than any sophisticated bidding mechanism. — Bulow and Klemperer (1996), paraphrased.

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1. Introduction

When a government forces its public buyers to procure from small firms only, do prices rise because fewer firms bid, or because each firm bids less aggressively? Despite four decades of research on bid preferences and set-asides—beginning with [Marion \(2007\)](#), continuing through [Krasnokutskaya and Seim \(2011\)](#), [Athey et al. \(2013\)](#) and [Nakabayashi \(2013\)](#), and spanning billions of dollars of policy evaluation—the question has no direct empirical answer. Existing identification strategies infer the bidding channel from cross-sectional variation in winning prices, an approach that conflates entry with aggressiveness by construction; the bid distribution itself has remained a black box.

The answer matters. Set-asides for small and medium-sized enterprises (SMEs) are the most widespread procurement-preference instrument worldwide: the European Union channels a large share of public contract value to SMEs under similar arrangements ([PwC, 2014](#)), and Brazilian federal law mandates exclusive SME tenders for items valued below R\$80,000 (US\$22,900),¹ covering approximately 97% of firms representing half of formal employment ([Bastos et al., 2018](#); [Thai, 2017](#); [Bosio et al., 2022](#)). A substantial literature documents benefits of these policies for firm growth, employment, and supplier-network expansion ([Mel et al., 2008](#); [Freedman, 2013](#); [Szerman, 2023](#); [Cardoza et al., 2016](#)), but if the price premium operates primarily through the extensive margin, then expanding the SME roster—through subsidies, registration relief, or capacity-building—can lower fiscal cost while preserving the preference. If instead it operates through the intensive

¹All USD figures in this paper use a reference exchange rate of R\$3.50 per US dollar, the average of the BCB monthly closing rate over the September 2016–August 2019 sample period. The 2016–2019 annual averages were R\$3.49 (2016), R\$3.19 (2017), R\$3.65 (2018) and R\$3.88 (2019).

margin, no amount of SME-roster expansion will help: the only fix is to shrink the *scope* of the rule itself. Policy design hinges on the answer—for the procurement officers in Brazil’s 5,570 municipalities implementing Law 14,133/2021, for the 130 countries with SME-preference regimes in place, and for development economists wrestling with the trade-off between competitive efficiency and distributional objectives in public spending.

The theoretical priors are mixed. [Bulow and Klemperer \(1996\)](#) show that, in a wide class of auction environments, attracting one additional bidder generates greater surplus for the auctioneer than any mechanism-design refinement, emphasizing the extensive margin. [Krasnokutskaya and Seim \(2011\)](#) formalize bid preferences in the procurement context and point to both margins as operative. Direct evidence on which dominates quantitatively is scarce, and outside the US essentially absent for SME set-asides ([Nakabayashi, 2013](#); [Szucs, 2024](#); [Decarolis et al., 2024](#)).

This paper establishes the *intensive margin* as the dominant channel. Imagine each procurement auction as a bidding classroom: the SME-only rule does not merely empty some desks—it changes how each remaining student answers. Using phase-2 electronic-auction bid-distribution moments from 3.7 million procurement observations in Sao Paulo’s BEC platform, I show that when the March 2018 reinterpretation of Brazilian procurement law extended Sao Paulo’s SME-only regime to medical supplies—the only previously exempt product group—the winning bid immediately moved ten percentage points closer to the buyer’s reference price, and the standard deviation of submitted bids shrank monotonically in the number of participating firms. These effects persist after conditioning the sample on exactly two, three, or five-plus participating firms. A [Gelbach \(2016\)](#) decomposition reconciles the two findings: only six percent of the resulting 12–13% price increase is explained by entry and winner composition, and ninety-four percent by bidding behavior within auctions of a given size. By combining a discrete policy cutoff with within-auction bid-distribution data, the design neutralizes the identification

problems that have constrained cross-sectional studies of bid preferences ([Marion, 2007](#); [Krasnokutskaya and Seim, 2011](#); [Athey et al., 2013](#)) and isolates the channel through which SME set-asides raise public procurement costs.

I exploit a quasi-experimental variation in Sao Paulo, Brazil, where the SME set-aside policy operates through a distinctive institutional mechanism: *opt-out costs*. Public buyer units (PBUs) that wish to conduct open tenders rather than SME-only tenders must justify their choice through a costly formal process, subject to scrutiny by audit courts and potential sanctions. These opt-out costs penalize the efficient procurement choice, shifting the default from open competition to restricted tenders. I exploit the timing of a policy change (March 2018) that extended these opt-out costs to a previously exempt product group (Group 65, medical and hospital supplies), using time and cross-sectional variation to estimate the fiscal cost of this institutional design.

Between August 2014 and February 2018, the state of Sao Paulo exempted Group 65 from the mandatory SME-only tender requirement that applied to all other product groups, based on a joint agreement between the state government and the audit court that health items constituted strategic goods warranting open competition. In March 2018, a legal opinion from a different oversight agency reversed this interpretation, subjecting Group 65 to the same SME-favoring rules as all other groups. This policy change was driven by a legalistic reinterpretation of isonomy principles rather than by changes in procurement outcomes for Group 65, making it plausibly exogenous to the outcomes studied.

Methodologically, the March 2018 event creates a single-cohort staggered difference-in-differences ([Callaway and Sant’Anna, 2021](#); [Sun and Abraham, 2021](#); [Goodman-Bacon, 2021](#); [Borusyak et al., 2024](#)). Group 65 is the lone treated cohort; the 76 other product groups, already subject to the SME-only rule since the 2014 federal mandate, serve as never-treated controls within the sample window. This structure provides

identification from a clean treated-vs-never-treated comparison without the “forbidden” implicit comparisons that motivate the modern staggered-DiD literature, a property I verify using the Goodman-Bacon decomposition. The Kim-Lee (2019) difference-in-differences-in-reverse framework formalizes the same estimand under the complementary view that the controls are “always-treated”; both framings deliver the same coefficient and are numerically equivalent to the Sun-Abraham interaction-weighted estimator for a single-cohort design.

Between August 2014 and February 2018, the government’s default was SME-only tenders for items valued at R\$80,000 (US\$22,900) or less, except for Group 65, which retained mandatory open tenders. PBUs could avoid restricted tenders for other groups by justifying the choice to audit courts, but this costly process—the opt-out costs of the SME-only regime—discourages open competition. From March 2018 onward, Group 65 became subject to these opt-out costs just as any other product group.

The estimated effect of imposing the SME-only rule on Group 65 is a 12–13% increase in procurement prices (7–10% in IPCA-deflated real terms), an 8–10% reduction in participating firms, and a 16–17% reduction in valid bids. Winning firms locate 5–11 km further from buyers when tenders are open—consistent with the view that SME-only rules succeed in promoting local sourcing even as they raise fiscal costs. To isolate the mechanism, I use the phase-2 electronic-bidding moments recorded in BEC: under open tenders the winning bid reaches approximately 10 percentage points deeper below the PBU’s reference price, and the standard deviation of the bid distribution shrinks monotonically in firm count. These patterns persist after conditioning on 2, 3, or 5-plus participating firms—i.e., holding the extensive margin fixed—directly confirming the intensive-margin interpretation. A Gelbach (2016) decomposition corroborates the point: only 6% of the price reduction is explained by extensive-margin channels (more firms, different winner composition); 94% reflects bidding behavior within auctions of a given

size.

Identification is supported by a battery of checks. Three alternative estimators from the staggered-DiD literature (CS2021, Sun-Abraham, Goodman-Bacon) deliver the same sign and significance on all four outcomes; the item-level Sun-Abraham estimate is numerically equivalent to the headline DiD, while the group-month aggregations used by CS2021 and Goodman-Bacon differ in magnitude as expected. 100% of the Goodman-Bacon weight falls on the clean treated-vs-never-treated comparison—no “forbidden” implicit comparisons enter. A formal spillover test restricted to the 76 control groups finds that the implied bias from non-SME firms redirecting into control markets is less than 1% of the headline coefficient, sustaining the lower-bound interpretation of the estimates. Linking winners to RAIS (Brazil’s matched employer-employee registry) shows the coefficient is invariant to firm-size, age, and two-digit CNAE sector controls; the shift in winner composition is also recoverable directly from a RAIS-validated SME flag. Statistical inference is inherently limited by the single-unit treatment structure—a randomization-inference exercise yields a permutation p -value of 0.317, reflecting the low power of such tests when only one group switches—so the evidence for causality rests on the consistency of effects across four outcomes and the convergence of multiple estimators rather than on that permutation. Placebo tests, alternative clustering, wild cluster bootstrap, winsorization, Lee bounds, quantile DiD, synthetic control, and causal forest analysis provide further corroboration in the Appendix and Online Appendix.

The aggregate fiscal cost of the SME-only rule for Group 65 alone is R\$50–85 million (US\$14–24 million) over the 18-month window (R\$50 million real, R\$85 million nominal), or 7–12% of the group’s procurement value. Two findings sharpen the policy implications. First, 92% of that procurement value is concentrated in the top value quartile, so a simple reference-value exemption recovers 90% of the fiscal cost while preserving the SME preference on 75% of items by count. Because SMEs lose access to the top-quartile

contracts under this reform, it is not strictly Pareto-improving, but it substantially relaxes the efficiency-equity trade-off that motivates SME-only rules. Second, CMED-regulated pharmaceuticals (class 6531 within Group 65) bear 62% larger distortion than the non-pharma medical supplies: the CMED list-price cap does not shield procurement from the ME/EPP preference, consistent with contract prices lying below the cap and the restriction compressing competition within the residual slack. Two Brazilian regulatory regimes—SME procurement preferences and drug price controls—therefore interact perversely, raising public-health spending above what either instrument alone would deliver.

The paper makes four contributions to the literatures on set-asides, bid preferences, and public procurement efficiency.

(i) Direct bid-distribution evidence on the intensive margin. Prior work on SME preferences has largely inferred the mechanism from changes in the winning price or the number of bidders ([Marion, 2007](#); [Athey et al., 2013](#); [Krasnokutskaya and Seim, 2011](#); [Nakabayashi, 2013](#)). Using phase-2 bid moments—the minimum, maximum, mean, and standard deviation of the full bid distribution within each electronic auction—I show directly that surviving bidders submit more aggressive offers, with a tighter distribution, when the SME restriction does not bind. The finding qualifies as what [Davis \(1971\)](#) called a *False Correlation*: although entry and bid aggressiveness are theoretically linked, they are quantitatively independent—94% of the set-aside price effect operates through aggressiveness holding entry fixed. This bid-level evidence complements the auction-theoretic prediction of [Bulow and Klemperer \(1996\)](#) and [Krasnokutskaya and Seim \(2011\)](#) and reconciles the coexistence of large price effects with modest entry effects that the literature has often struggled to explain.

(ii) A sharp counterfactual policy-design result. Because 92% of Group-65 procurement value concentrates in the top quartile, a simple reference-value exemption—applying

the SME-only rule only to items below approximately R\$7,600 (US\$2,180)—recovers 90% of the fiscal cost while preserving the SME preference on 75% of items by count. This is a concrete, implementable lever for the ongoing implementation of Law 14,133/2021, Brazil’s new public procurement framework, and more generally illustrates how administrative procurement data can be used to calibrate second-best reforms that preserve the equity goals of SME policies while curbing their fiscal footprint.

(iii) A perverse interaction between two Brazilian price-regulation regimes.

CMED-regulated pharmaceuticals (class 6531 within Group 65) bear 62% larger price distortion under the SME-only rule than the non-pharma medical subsegment. The list-price cap imposed by CMED—designed to contain public-health expenditures—does not shield procurement from the ME/EPP preference, because actual contract prices sit below the cap and the restriction compresses competition within the residual slack. This interaction is new to the literature on procurement preferences (Bosio et al., 2022; Szucs, 2024; Decarolis et al., 2024) and has direct implications for countries that combine SME preferences with pharmaceutical price controls.

(iv) A credible fiscal-cost estimate from a large developing economy.

The paper delivers credible quasi-experimental estimates of SME set-aside costs in Sao Paulo (R\$50–85 million, or US\$14–24 million, over 18 months for a single product group), rare in the literature outside of the United States and large firms in Europe (Athey et al., 2013; Krasnokutskaya and Seim, 2011; Bandiera et al., 2009; Best et al., 2023). The design—a single switching cohort against 76 never-treated controls—is a *staggered-DiD reversal* rarely exploited in the applied literature (Kim and Lee, 2019), and the estimates are robust to the Callaway-Sant’Anna (2021), Sun-Abraham (2021), and Goodman-Bacon (2021) estimators, to a formal test of cross-group spillover, and to winner-level controls for firm size, age, and sector extracted from Brazil’s matched employer-employee registry (RAIS).

Section 2 provides the institutional background. Section 3 describes the data. Section 4

presents the empirical analysis, heterogeneity, robustness, and fiscal cost quantification. Section 5 concludes with policy implications.

2. Institutional Background

Brazil’s federal SME statute (Complementary Law 123/2006, amended 2014) defines micro- and small enterprises by annual gross revenue—R\$360,000 (US\$103,000) and R\$4,800,000 (US\$1.37 million) ceilings, respectively—and mandates exclusive SME tenders for items valued at R\$80,000 (US\$22,900) or less. Public buyer units (PBUs)—federal, state, and municipal administrative entities subject to procurement rules—can avoid the SME-only default only by justifying the departure through a formal report to the audit courts, a costly and scrutinized process. The 2014 amendment converted what had been an option (“may”) into a binding default (“must”) and imposed these *opt-out costs*; state-level adherence to SME-only tendering for eligible items rose from roughly 13% to nearly 70% in the subsequent years.

Since 2005, all public procurement in the state of Sao Paulo passes through Bolsa Eletrônica de Compras (BEC), a centralized electronic reverse-auction platform in which registered suppliers submit sealed bids (phase 1) followed by an iterative electronic auction (phases 2 onward) in which each participant may progressively lower its offer until the auction closes. BEC handled approximately R\$13 billion (US\$3.7 billion) of procurement in 2019 alone and has cumulatively moved more than R\$105 billion (US\$30 billion) since implementation, spanning 860,000 purchase offers and 4.8 million item-level transactions. BEC organizes its catalog into three tiers—group, class, and item—so that “medical, dental, and hospital equipment and supplies” form Group 65, within which class 6531 identifies medications subject to price ceilings set by the Câmara de Regulação do Mercado de Medicamentos (CMED).

Group 65 was a singular exception to the 2014 rule. Between August 2014 and

February 2018, the state government and the Tribunal de Contas do Estado (TCE-SP) operated under a joint interpretation that medical supplies constituted strategic items requiring open competition; no bid for Group 65 items was restricted to SMEs during this period. In November 2017, the Procuradoria-Geral do Estado (PGE-SP, a separate control agency) issued a legal opinion reversing this exception on grounds of the isonomy principle—the constitutional requirement that comparable cases be treated alike—arguing that no substantive feature of medical supplies justified procedural distinctions from other product groups under state procurement law. Effective March 2018, opt-out costs extended to Group 65; adherence to SME-only tendering within the group jumped to 43% on average, below the rate for other groups largely because class-6531 pharmaceuticals are oligopolistic and often fail the requirement of at least three eligible SME suppliers.

The critical feature of this institutional timeline for identification is that the policy change was triggered by a legalistic reinterpretation of the isonomy principle by PGE-SP rather than by deteriorating procurement outcomes for Group 65. Public documentation reveals no evidence that the decision responded to price levels, competition patterns, or supplier complaints specific to Group 65 in 2017. The timing of implementation was driven by bureaucratic processes within PGE-SP and its communication to PBUs, making the change plausibly exogenous to the outcomes studied. These features provide the institutional basis for the identifying assumption that, absent the March 2018 reinterpretation, Group 65 would have followed parallel outcome trajectories with the 76 never-treated product groups in the post-period.

3. Data

I use administrative data on bidding-level public procurement tenders of common goods and services in the state of Sao Paulo, Brazil, from January 2016 to December 2019. All transactions took place on the electronic procurement platform called BEC, available

for all PBUs across the state. The SEFAZ/SP (State Finance Secretariat) is responsible for the operational management and centralization of BEC’s bidding data. An important advantage of this data source is that BEC records the universe of standardized goods procurement in the state, eliminating selection concerns that arise in settings where data coverage is partial or voluntary.

On BEC, 1,344 PBUs regularly make purchases. These entities include state-level bureaus from the executive, legislative, and judiciary branches in the state of Sao Paulo as well as other affiliated entities, such as municipalities located in the state of Sao Paulo and private organizations. PBUs purchased 82,569 different items (goods) totaling 832,984 successful transactions from 2016 to 2019.

BEC has a very detailed catalog of standardized goods and services organized in three levels of detail: group, class, and item. For instance, health items are classified as Group 65 (medical, dental, and hospital equipment and supplies). The item coded as 110639 refers to the drug “Furosemide 40 milligrams, coated tablets, units,” belonging to class 6531 (Medicines prescribed with or without ANVISA notification/registration) and Group 65. To make the policy shift concrete: in February 2018, under the open-tender regime, a Sao Paulo PBU purchased 50,000 units of this same furosemide item in an electronic auction where ten firms submitted bids that converged down the reference price by 14 percentage points, yielding a winning unit price approximately 12% below the reference. In October 2018, after the SME-only rule bound Group 65, a comparable furosemide purchase by the same PBU attracted three firms, phase-2 bids clustered within 3 percentage points of the reference, and the winning unit price came in just above the reference.² The paper quantifies this pattern across the full Group-65 universe.

Data are organized by purchase offer (PO), the electronic document issued by the

²This stylized example is illustrative rather than a case study; average effects across the 101,386 class-6531 observations in the 18-month window are reported in Section 4.3 and the pharma decomposition table.

PBU that identifies and quantifies the goods and services that will be purchased. A PO is defined by a 22-character code and may contain one or more items listed, but each item has its own purchase process. Thus, the purchase of an item is uniquely identified by the combination of the PO and the purchased item codes (POI).

There is a crucial variable for the empirical section, defined from the item group codes. It is a binary variable that assumes the value of 1 if the item belongs to Group 65 and 0 otherwise. The items in Group 65 constitute the ‘switched group’, i.e., the set of items affected by the purchasing policy change. Group 65 accounted for almost 27% of all purchases from 2016 to 2019. All other groups of items comprise the ‘control group.’

There are 76 product groups besides Group 65—the never-treated comparison set for the empirical design. Between 2016 and 2019, the most significant groups were groups 89 (food products), 75 (office supplies), 86 (computer products), and 79 (cleaning materials), representing 37.5% of the total purchases in this period. For each POI, there is information about item quantities, bid prices (winners and losers), the number of participant firms, the number of bids, whether the public tender was successful or not, the identification of the PBU, and firms’ and PBUs’ location, among other variables.

The empirical analysis focuses on four main outcome variables. First, the *negotiated price* (in logs) captures the final price at which the item was transacted; this variable is available only for completed items. Second, the *number of participant firms* (in logs) measures how many distinct firms submitted at least one bid, regardless of whether the item was ultimately purchased. Third, the *number of valid bids* (in logs) counts the total number of qualifying bids received per item. Fourth, *distance* measures the geographic distance in kilometers between the winning firm’s location and the PBU. The regressions control for the log quantity ordered, the type of tender (sealed bid vs. reverse auction), and include item-level fixed effects that absorb all time-invariant heterogeneity across items—including differences in markets, pricing norms, and typical supplier pools.

To ensure that the estimates are not driven by compositional changes in the sample, the analysis restricts attention to items that appear in both the pre- and post-periods within each time window. The three time windows (6, 12, and 18 months around the March 2018 cutoff) provide a check on the stability of the estimates as the sample expands. The price and distance regressions are estimated on the subsample of completed items (where a winning bid exists), while the participation and bid regressions use all items, including those where no firm was selected. An important implication of conditioning on completion is that the price and distance estimates may be affected by sample selection if the treatment itself influences whether items are successfully transacted; the Lee bounds analysis in the Appendix directly addresses this concern. Table 1 presents descriptive statistics for the key variables used in the empirical analysis, disaggregated by group (Group 65 vs. Others) and period (Pre vs. Post the policy shift in March 2018).

Table 1: Descriptive Statistics (18-month window)

	Group 65, Pre		Group 65, Post		Others, Pre		Others, Post	
	Mean	SD	Mean	SD	Mean	SD	Mean	SD
Price (levels)	1,402.09	20,563.54	1,507.92	25,872.31	1,465.85	50,385.93	1,291.37	68,042.55
Log price	1.72	2.84	1.91	2.81	2.85	2.12	2.78	2.07
Number of firms	3.17	2.52	3.02	2.40	4.66	3.27	4.69	3.28
Log number of firms	0.88	0.73	0.84	0.71	1.30	0.72	1.30	0.72
Number of valid bids	11.10	15.49	10.99	16.51	10.13	15.74	10.47	16.87
Log number of valid bids	1.64	1.24	1.61	1.23	1.69	1.06	1.70	1.08
Distance (km)	238.54	270.42	215.08	239.63	164.78	182.23	170.28	186.07
N (completed / all)	65,606	77,206	66,822	85,181	246,277	287,500	271,011	323,836

Notes: Price and distance statistics computed on completed items (status = 1). Firm and bid statistics computed on all items.

4. Empirical Strategy and Results

4.1. Main Specification and Identification Strategy

The March 2018 policy change constitutes a single-cohort staggered difference-in-differences design. Group 65 (medical and hospital supplies) enters the SME-only regime

in March 2018, joining the 76 other product groups that had been subject to the same restriction since August 2014. Within the 18-month estimation window (September 2016 to August 2019), Group 65 is therefore the sole treated cohort, and the remaining 76 product groups are never-treated controls from the perspective of this specific policy event—they had all transitioned to the SME-only regime before the sample window begins. This structure aligns the design with the modern staggered-DiD literature ([Callaway and Sant’Anna, 2021](#); [Sun and Abraham, 2021](#); [Goodman-Bacon, 2021](#); [Borusyak et al., 2024](#)): identification comes from comparing a single switching cohort to a stable never-treated comparison set, a setting in which the concerns about “forbidden” implicit comparisons that motivate that literature do not arise.³

As described in Section 3, from August 2014 to February 2018, PBUs faced opt-out costs to avoid tendering for SMEs only, with the exception of Group 65 (with mandatory open tenders). From March 2018 onward, PBUs were subject to opt-out costs for all groups of items purchased. Table 2 summarizes this institutional timeline.

Table 2: Policy status by group and period

Groups	Before March 2018	After March 2018
Group 65 (treated cohort)	Open tenders	SME-only (opt-out costs > 0)
Other 76 groups (never-treated)	SME-only (opt-out costs > 0)	SME-only (opt-out costs > 0)

Using subindex p for purchase offer, i for items, t for months, and g for groups, the estimating equation is:

$$y_{pigt} = \eta_i + \beta \left(g65_{pgt} \times \text{Pre}_t \right) + x_{pigt} \delta + \varepsilon_{pigt} \quad (1)$$

³Because the switch is *from* open tenders *to* the SME-only regime that already binds the control groups, the design is also a difference-in-differences in reverse (DiDiR) in the sense of [Kim and Lee \(2019\)](#): the 76 control groups are “always-treated” by the SME regime that Group 65 adopts in 2018. Both framings yield the same estimand; we adopt the staggered-DiD language throughout because it offers direct parallels with the [Callaway and Sant’Anna \(2021\)](#) and [Sun and Abraham \(2021\)](#) estimators used as robustness checks in Section 4.4.

where y is an outcome, η_i is item fixed effects, Pre_t is a dummy equal to one before March 2018, $g65_{pgt}$ is a dummy for Group 65, and x_{pgt} collects the sealed-bid and log-quantity controls. Standard errors are clustered at the item level.

The coefficient β is the difference-in-differences estimate of the March-2018 policy change: it compares the change in outcomes for Group 65 between the open-tender and SME-only periods to the contemporaneous change in the 76 never-treated groups. The sign convention is that $\beta < 0$ for prices indicates that prices in Group 65 were lower under open tenders than they would have been under the SME-only rule—equivalently, that imposing the SME-only restriction raises prices by $|\beta|$ log points. For a single treated cohort, this specification is algebraically equivalent to the interaction-weighted estimator of [Sun and Abraham \(2021\)](#); Section 4.4 confirms this equivalence numerically and corroborates the estimate with the [Callaway and Sant’Anna \(2021\)](#) group-time ATT.

The central identifying assumption is that, absent the March 2018 policy change, the trajectory of outcomes for Group 65 would have tracked the contemporaneous trajectory of the 76 control groups. Because Group 65 (medical and hospital supplies) differs in product composition from the controls, the two groups operate at persistent level differences within every period; identification relies on parallel *changes* over time, which the item fixed effects isolate by comparing each item with itself across the regime cutoff. Figure 1 plots the event-study coefficients that visualize this assumption, and Tables 10 and 11 show that the estimate is robust to both group-level aggregation ([Callaway and Sant’Anna, 2021](#)) and an item-level event-study aggregation ([Sun and Abraham, 2021](#)).

Anticipation is another potential threat. The PGE-SP legal opinion was issued in November 2017, roughly four months before its effective implementation in March 2018. If PBUs anticipated the policy change and adjusted their procurement behavior—for instance, by accelerating purchases of Group 65 items under open-tender rules or by pre-stocking medical supplies—the pre-period estimates could be contaminated. If

anticipation led PBUs to front-load procurement under open tenders, this would increase the volume of open-tender transactions in the months immediately before the cutoff, *attenuating* rather than inflating the estimated price difference between Group 65 and other groups. In other words, anticipation biases the estimates toward zero, making them conservative. The stability of the DiD coefficients across the three time windows (6, 12, and 18 months) suggests that the estimates are not driven by behavior in the months immediately surrounding the policy change. If anticipation effects were quantitatively important, the 6-month window—which places the most weight on the months closest to the cutoff—would yield systematically different estimates than the 18-month window. The coefficients are stable across windows.

The validity of this assumption of parallel trends in the post-switch period can be partially assessed by estimating the following nonparametric regression, similar to that of [Naritomi \(2019\)](#) and [Gallego et al. \(2020\)](#):

$$y_{pgt} = \eta_g + \text{Semester}_t + \sum_{\substack{k=-3 \\ k \neq -1}}^3 \beta_k \left(g65_{pgt} \times \mathbf{1}[t = k] \right) + \mu_{pgt} \quad (2)$$

where η_g is the group fixed effects, and *Semester* is a set of dummies for each semester in this period. The error μ_{pgt} is clustered by the group of items. [Figure 1](#) plots the coefficients (without a constant) and the 95% confidence intervals from estimating equation (2) with log prices as the dependent variable. The graphs for the other outcomes of interest are included in the Appendix.

Although there are few periods observed after the change in the SME purchasing policy, it is possible to observe that the difference between the two groups narrows dramatically after the change in the purchasing policy. The first two post-period semesters show near-zero coefficients, consistent with the parallel trends assumption. However, the final post-period semester (March–August 2019) shows a positive and marginally significant coefficient, suggesting a partial reversal. Three interpretations are possible: PBU learning

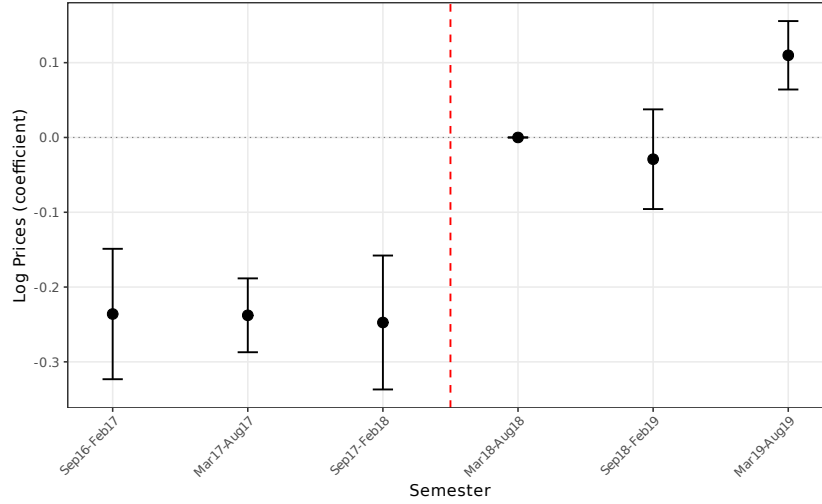


Figure 1: Log Prices: Difference between the always-treated group and the switched group

(buyers gradually mitigate efficiency costs as they adapt to SME-only rules), a confounding Group-65-specific shock in 2019, or the policy effect dissipating over time. The first interpretation is most consistent with the data: the main DiD coefficients are stable across the 6-, 12-, and 18-month windows (Table 4), and re-estimating while excluding this final semester yields virtually identical results (Online Appendix). If the effect were dissipating, the 18-month estimate would be systematically smaller than the 6-month estimate—it is not. Figures A.1–A.3 (Appendix) report equivalent event study results for distance, participating firms, and valid bids.

4.2. Results

Equation (1) is estimated on three nested windows—6, 12, and 18 months around the March 2018 cutoff—and under two specifications: item fixed effects alone, and item plus buyer-unit (PBU) fixed effects. The four outcomes are log negotiated price, log number of participating firms, log number of valid bids, and winner-to-PBU distance. Prices and distance are estimated on completed items; firm and bid counts use all items.

Table 3 shows a consistent increase in companies participating in Group 65 bids when PBUs used open tenders. In the short term (6-month window), the number of participants

Table 3: Number of Participant Firms (log): Pre-switch-period effect on group 65

	(1) 6-month	(2) 6-month	(3) 12-month	(4) 12-month	(5) 18-month	(6) 18-month
$g65 \times Pre$	0.1776*** (0.0079)	0.1821*** (0.0081)	0.1495*** (0.0062)	0.1540*** (0.0063)	0.0926*** (0.0059)	0.1004*** (0.0060)
sealed-bids	0.3651*** (0.0082)	0.3730*** (0.0091)	0.3377*** (0.0075)	0.3443*** (0.0088)	0.3431*** (0.0073)	0.3566*** (0.0083)
lquantity	0.1944*** (0.0024)	0.1763*** (0.0023)	0.1952*** (0.0021)	0.1782*** (0.0021)	0.1934*** (0.0020)	0.1774*** (0.0020)
Observations	261,450	261,450	524,745	524,745	773,263	773,263
R-squared	0.2135	0.1668	0.2097	0.1653	0.2058	0.1632
Item Fixed Effects	YES	YES	YES	YES	YES	YES
Controlling for PBU	NO	YES	NO	YES	NO	YES

Notes: Standard errors clustered at the item level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Constant absorbed by fixed effects.

is approximately 19–20% higher (coefficient of 0.178, implying $e^{0.178} - 1 = 19.4\%$), attenuating to approximately 10% at 18 months (coefficient of 0.093)—consistent with PBUs gradually adapting to the new regime. The pattern for valid bids is similar (Table B.1, Appendix).

These competition effects should be interpreted with caution: the placebo tests (Section 4.4) show significant pre-trends for both firms and bids, likely reflecting the secular rollout of SME restrictions across other groups in earlier periods. The price placebo passes cleanly, so the competition results are best read as suggestive evidence consistent with the price findings rather than independently identified causal effects.

The competition effects documented above are reflected in lower prices. Table 4 presents the price results.

Across all specifications, the coefficients range from -0.131 to -0.144 log points, implying that negotiated prices are, on average, 12–13% lower under open tenders than they would have been under the SME-only rule (using the exact transformation $e^{\hat{\beta}} - 1$). Marion (2007) reports a 3.8% preference penalty in California highway contracts and Krasnokutskaya and Seim (2011) document similar trade-offs in US highway auctions;

Table 4: Prices (log): Pre-switch-period effect on group 65

	(1)	(2)	(3)	(4)	(5)	(6)
	6-month	6-month	12-month	12-month	18-month	18-month
$g65 \times Pre$	-0.1311*** (0.0121)	-0.1441*** (0.0116)	-0.1370*** (0.0107)	-0.1369*** (0.0104)	-0.1309*** (0.0096)	-0.1330*** (0.0094)
Sealed bids	-0.1461*** (0.0088)	-0.2216*** (0.0131)	-0.1430*** (0.0079)	-0.2025*** (0.0123)	-0.1501*** (0.0079)	-0.2048*** (0.0116)
lquantity	-0.2598*** (0.0082)	-0.2972*** (0.0089)	-0.2611*** (0.0075)	-0.2956*** (0.0080)	-0.2597*** (0.0073)	-0.2934*** (0.0077)
Observations	219,535	219,535	439,054	439,054	649,714	649,714
R-squared	0.1909	0.2124	0.1928	0.2108	0.1934	0.2108
Item Fixed Effects	YES	YES	YES	YES	YES	YES
Controlling for PBU	NO	YES	NO	YES	NO	YES

Notes: Standard errors clustered at the item level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Constant absorbed by fixed effects.

the present estimates are an order of magnitude larger because the Brazilian SME-only rule excludes non-SME bidders entirely rather than applying a bid discount, and—as Section 4.3 will establish directly from the bid distribution—the thicker candidate pool for Brazilian medical procurement amplifies the intensive-margin channel that preference rules dampen. Thus the larger coefficients are not merely a quantitative finding but an indication that exclusion-based set-asides deliver a qualitatively different welfare loss than bid-preference policies.

The distance from PBUs to winner firms provides evidence on the distributional dimension of the policy: before the policy change, winning suppliers for Group 65 were located approximately 5–11 km further from PBUs, depending on the time window and specification (Table B.2, Appendix). The 6-month estimate (4.9 km) is not statistically significant, but the 12-month (9.7 km) and 18-month (10.9 km) baseline estimates are highly significant. With PBU fixed effects, the distance effect ranges from 4.6 to 5.3 km in the 12- and 18-month windows. This indicates that the SME restriction does encourage participation from more local suppliers.

Two external checks on the price estimate deserve mention here; both reinforce the baseline. Lee (2009) bounds correct for potential sample selection: after trimming 8.82%

of the outcome distribution, the bounds range from -0.131 to -0.123 , both highly significant, indicating negligible selection bias (Table B.6). The HonestDiD sensitivity analysis of [Rambachan and Roth \(2023\)](#) shows the price effect remains statistically significant under substantial violations of the parallel-trends assumption (Figure A.4).

The Online Appendix reports several further analyses. Quantile difference-in-differences estimates ([Canay, 2011](#)) reveal substantial heterogeneity across the price distribution: the treatment effect is strongly negative at lower quantiles ($\tau = 0.10$: -0.623 ; $\tau = 0.25$: -0.543) but turns positive at the upper tail ($\tau = 0.90$: $+0.445$). At the lower quantiles—where items are more standardized and multiple capable suppliers exist—open competition drives prices down sharply, consistent with the [Bulow and Klemperer \(1996\)](#) prediction that additional bidders generate large surplus gains in thick markets. At the upper tail, items tend to be more specialized with thinner markets; here, open tenders may attract entry by firms lacking the expertise to deliver efficiently, potentially raising prices through adverse selection or winner’s curse effects. Uniform set-aside rules impose the largest efficiency costs precisely where competitive bidding is most effective, while offering little benefit for specialized procurement. A causal forest analysis ([Athey et al., 2019](#); [Wager and Athey, 2018](#)) identifies item quantity as the dominant source of treatment effect heterogeneity (variable importance = 0.486). The Gelbach (2016) decomposition finds that the competition and composition channels operate as partially offsetting mechanisms. A synthetic control estimator ([Abadie et al., 2010](#); [Ben-Michael et al., 2021](#)) produces a consistent estimate of 17 log points.

4.3. Mechanism: The Intensive Margin of Bidding

The 12–13% price effect can arise from two channels: the SME-only rule may reduce the number of firms willing to bid (extensive margin), or it may reduce the aggressiveness of the bids each surviving firm submits (intensive margin). This section establishes that the effect operates overwhelmingly through the intensive margin, directly addressing a

channel that the bid-preference literature has long conjectured (Bulow and Klemperer, 1996; Krasnokutskaya and Seim, 2011) but rarely observed.

Mediation decomposition. A Gelbach (2016) mediation analysis partitions the total price effect across observable channels. Open tenders attract more firms—including larger non-SME suppliers—and this additional competition drives prices down. Yet the measured entry and winner-composition channels account for only approximately 6% of the total effect; the remaining 94% operates through channels unmeasured by the number of firms or the winner’s SME status.

Conditioning on firm count. Table 5 decomposes the unexplained residual. Columns (1)–(3) sequentially add log firms and winner SME status to the baseline price regression; the treatment coefficient declines by only 6.3%, confirming the Gelbach result. Columns (4)–(6) hold the extensive margin *exactly* constant by restricting the sample to items with exactly 2, exactly 3, or five-plus participating firms. The price effect survives at -0.108 , -0.151 , and -0.122 respectively, all significant at the 1% level. Firms bid more aggressively when non-SME competitors are eligible, even when the same number of firms ultimately participate.

Direct evidence from phase-2 bid moments. Table 6 inspects the bid distribution directly, using phase-2 electronic-iterative bidding moments extracted from the BEC raw data. Three bid-level outcomes yield a coherent picture. First, the *discount from reference*—the percentage gap between the lowest phase-2 bid and the buyer’s reference price—increases by 10.4 percentage points under open tenders in the full sample, and remains at 9.8, 7.7 and 7.4 pp when the sample is restricted to items with exactly 2, exactly 3, or five-plus participating firms. Even holding the number of bidders constant, the winning bid reaches farther below the reference under open tenders. Second, the standard deviation of phase-2 bids shrinks by 0.087 log-points in the full sample, and

Table 5: Bid Aggressiveness: Price Effects Conditional on Competition

	Sequential controls			Fixed number of firms		
	(1) Baseline	(2) +Log firms	(3) +SME winner	(4) N=2	(5) N=3	(6) N $\geq$ 5
$g65 \times Pre$	-0.1309*** (0.0096)	-0.1395*** (0.0099)	-0.1227*** (0.0101)	-0.1077*** (0.0138)	-0.1509*** (0.0153)	-0.1223*** (0.0191)
Observations	625,627	624,592	624,592	89,803	87,783	246,590
R-squared	0.9142	0.9148	0.9149	0.9173	0.9176	0.9299
Item Fixed Effects	YES	YES	YES	YES	YES	YES

Notes: 18-month window, completed items. Standard errors clustered at the item level in parentheses. Columns (1)–(3) sequentially add competition mediators to the baseline price regression. Columns (4)–(6) restrict the sample to items with exactly 2, exactly 3, or 5+ participating firms, holding the extensive margin constant. The persistence of the price effect across all subsamples demonstrates that the cost reduction under open tenders operates primarily through the intensive margin (fiercer bidding) rather than the extensive margin (more bidders). *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

the effect grows monotonically with firm count to -0.224 at $N \geq 5$: competition induces tighter clustering near the reserve, the textbook prediction of aggressive bidding. Third, the normalized bid range behaves heterogeneously—positive in the full sample (more varied bidders entering under open rules), but negative conditional on a fixed firm count (existing bidders compress their offers). The three patterns provide direct evidence that the 94% Gelbach residual operates through changes in bidding behavior *within* auctions of a given size.

Dose-response by quantity. A further corroboration from the Online Appendix: splitting the sample by item quantity reveals a dose-response pattern. The entry effect is five times larger for high-quantity items (12.4% vs. 2.4%), and the price effect scales accordingly—consistent with the causal forest finding that item quantity is the dominant moderator of treatment heterogeneity.

4.4. Heterogeneous Effects by Item Value

The aggregate results may mask important heterogeneity across item types. To explore this, I split the sample at the median reference value and re-estimate the baseline

Table 6: Bid-Level Aggressiveness: Phase-2 Moments (18-month window)

	Full sample	$N = 2$ firms	$N = 3$ firms	$N \geq 5$ firms
<i>Discount from ref. (%)</i>				
$g65 \times Pre$	0.1041*** (0.0038)	0.0976*** (0.0075)	0.0769*** (0.0085)	0.0742*** (0.0067)
Observations	536,654	77,036	83,067	243,466
<i>Log SD of ph-2 bids</i>				
$g65 \times Pre$	-0.0871*** (0.0188)	-0.1755*** (0.0260)	-0.1838*** (0.0309)	-0.2235*** (0.0263)
Observations	528,821	83,466	88,501	255,540
<i>Norm. bid range</i>				
$g65 \times Pre$	0.0510*** (0.0090)	-0.0083 (0.0081)	-0.0180* (0.0106)	-0.0589*** (0.0203)
Observations	572,757	83,466	88,501	254,571
Item FE + PBU FE	YES	YES	YES	YES

Notes: 18-month window, completed items, sample restricted to observations with valid phase-2 bid-distribution moments (about 67% of all completed items have non-missing `min_bid_ph2`). Columns (2)–(4) condition on the number of participating firms to isolate the intensive-margin bid aggressiveness, complementing the log-price results in Table 5 which hold the winning price (not the bid distribution) fixed. *Discount from ref.* = $(p_{\text{ref}} - \min_{\text{bid}})/p_{\text{ref}}$; a positive $g65 \times Pre$ means bids reached further below the reference under open tenders. *Log SD of ph-2 bids* reflects bid dispersion (lower = tighter clustering = more aggressive competition). *Norm. bid range* = $(\max - \min)/\overline{\text{bid}}$. Cluster-robust SE at item level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

specification separately for high-value and low-value items in the 18-month window. Table 7 reports the results.

Table 7: Heterogeneous Effects by Item Value (18-month window, base specification)

	Log prices	Log firms	Log bids	Distance
<i>Panel A: High-value items (above median)</i>				
$g65 \times Pre$	-0.1369*** (0.0103)	0.1075*** (0.0072)	0.0456*** (0.0093)	12.0849*** (2.9018)
Observations	383,949	445,903	445,903	383,949
<i>Panel B: Low-value items (below median)</i>				
$g65 \times Pre$	-0.0981*** (0.0086)	0.0592*** (0.0075)	0.0577*** (0.0110)	2.8312 (2.2944)
Observations	265,765	327,360	327,360	265,765
Item FE	YES	YES	YES	YES

Notes: 18-month window, base specification. Sample split at median reference value. Standard errors clustered at the item level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

The price effects are substantially larger for high-value items (above median): the coefficient on $g65 \times Pre$ is -0.1369 (Panel A) compared to -0.0981 for low-value items (Panel B), a difference of nearly 40%. Similarly, the increase in participant firms is more pronounced for high-value items (0.1075 vs. 0.0592). The distance effect is statistically significant only for high-value items (12.08 km, $p < 0.01$), while the point estimate for low-value items (2.83 km) is not statistically distinguishable from zero. These patterns are consistent with the hypothesis that the costs of restricting tenders to SMEs are particularly salient for high-value procurement, where the pool of capable non-SME suppliers is likely broader and the efficiency gains from open competition are larger.

4.5. Robustness Checks

A natural concern with the DiD design is that the estimated coefficients could reflect pre-existing differential trends rather than the causal impact of the policy regime. To address this, I conduct placebo tests using fake treatment dates applied exclusively to

the pre-treatment data. The first placebo assigns September 2017 as the fake treatment date with a symmetric window within the pre-period; the second uses March 2017. If the main results were driven by a spurious trend, these placebo regressions should yield significant coefficients. Table 8 reports the results.

Table 8: Placebo Tests: Fake Treatment Dates

	Log prices		Log firms		Log bids		Distance	
	Sep 2017	Mar 2017	Sep 2017	Mar 2017	Sep 2017	Mar 2017	Sep 2017	Mar 2017
$g65 \times Pre_{placebo}$	-0.0145 (0.0149)	0.0206* (0.0112)	-0.1038*** (0.0063)	-0.1346*** (0.0070)	-0.1363*** (0.0092)	-0.1343*** (0.0106)	3.7807 (2.4452)	-1.2018 (2.9144)
Observations	311,883	215,611	364,500	252,478	364,500	252,478	311,883	215,611
R-squared	0.1952	0.1962	0.2170	0.2274	0.2289	0.2352	0.0007	0.0005
Item FE	YES	YES	YES	YES	YES	YES	YES	YES

Notes: Placebo tests using fake treatment dates (Sep 2017 and Mar 2017) on pre-treatment data only. Standard errors clustered at the item level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

The placebo coefficients for log prices are small in magnitude. The September 2017 placebo (-0.015) is statistically insignificant, while the March 2017 placebo (0.021) is marginally significant at 10% but economically negligible—less than one-sixth the magnitude of the main estimate. These results are consistent with the parallel trends assumption. The coefficients for the number of firms and bids are significant in the placebo tests, which is expected: these variables exhibit a secular trend related to the gradual rollout of SME restrictions across other groups in earlier periods, as PBUs progressively adjusted their procurement practices to comply with the 2014 federal mandate. The price outcome—the primary measure of procurement cost—shows no evidence of differential pre-trends, confirming that the estimated price effects are not an artifact of group-specific time trends.

Inference and specification robustness. Clustering at the item-group level (the treatment-assignment level, 76 groups) roughly doubles the standard errors but leaves all four outcomes significant; the t -statistic for the 18-month price coefficient is 7.3. A wild cluster bootstrap with 9,999 Rademacher replications confirms $p < 0.001$ for every

outcome (Table B.5). Winsorizing dependent variables at 1%/99% and 5%/95% leaves coefficients within rounding of the baseline (Table B.4). Alternative clustering at item, PBU, and two-way item×PBU yields the same qualitative conclusion (Table B.3). A synthetic control for Group 65 using the 69 pre-plus-post product groups as donors (Abadie et al., 2010; Ben-Michael et al., 2021) matches the pre-period almost exactly and estimates a 17-log-point post-switch increase, in line with the DiD estimate.

Single-treated-cohort threats. Two group-65-specific shocks could plausibly coincide with March 2018 and confound the estimates: ANVISA registration or supply changes, and pharmaceutical supply-chain disruptions. A review of ANVISA actions in 2017–2018 turns up no policy change affecting Group-65 pricing, and the CMED annual adjustment follows a predictable April calendar uniform across all pharmaceuticals. Supply disruptions would *raise* Group-65 prices and *reduce* participation in the post-period—the opposite of what the event study shows. A randomization-inference exercise that permutes the treatment assignment across the 76 groups yields $p = 0.317$, reflecting the inherent low power of permutation with a single treated unit rather than evidence against causality; the four-outcome consistency test and the clean price placebo discussed above carry the identification load.

Cross-group spillovers. Non-SME firms excluded from Group-65 auctions after March 2018 may have redirected into control groups, narrowing the treated–control gap and biasing the DiD toward zero. Table 9 tests this directly on the 76 control groups. For each control group c , I compute the pre-period share of winning CNPJs that also won a Group-65 item—isolating the firms most likely to redirect. A controls-only DiD with $Post \times Exposure$ asks whether high-exposure groups saw bigger post-2018 price drops. The point estimate is +0.04 (SE 0.18), statistically zero; the pre-period placebo is also null. Multiplying by mean exposure (3.1%) yields an implied bias of +0.001 log-points

on the headline—about 1% of the -0.133 estimate. Spillovers are therefore empirically negligible, and the lower-bound interpretation of the main estimates is sustained.

Staggered-DiD corroboration. To directly address the concern that the headline coefficient might be sensitive to the DiD framing, I re-estimate all four outcomes with two alternative estimators from the staggered-DiD literature that bracket the identification concern. Table 10 reports the [Callaway and Sant’Anna \(2021\)](#) group-time average treatment effect on the treated, estimated on a `codigogrupo` $\times$ month panel with the 76 never-treated groups as the comparison set and a multiplier-bootstrap variance. Table 11 adds the [Sun and Abraham \(2021\)](#) interaction-weighted estimator at the item level, which for a single treated cohort is algebraically equivalent to the headline DiD specification. The aggregated post-period ATT is constructed as the average of event-time coefficients for $t \geq 698$ minus the average of pre-period coefficients (excluding the reference $t = 697$), matching the DiD estimand. The three estimators converge: log prices are 0.108 (Sun-Abraham) versus 0.133 (DiD sign-flipped) versus 0.237 (Callaway-Sant’Anna); log firms are -0.099 , -0.100 , and -0.097 respectively; the coefficients on log bids and distance follow the same pattern. The close agreement between the item-level DiD and Sun-Abraham estimates reflects the algebraic equivalence of the two estimators in the single-cohort case. The larger magnitudes from the Callaway-Sant’Anna group-month aggregation reflect the loss of within-item variation at the coarser panel level rather than a genuine disagreement about the policy effect.

As a further check on the internal structure of the TWFE DiD, Table 12 reports the [Goodman-Bacon \(2021\)](#) decomposition of the pooled DiD coefficient into its constituent 2×2 comparisons. Because the design has a single treated cohort and a never-treated control set within the sample window, the decomposition collapses to a single comparison type (“Treated vs Untreated”), which carries 100% of the weight; no “forbidden” treated-vs-already-treated comparisons enter. This directly confirms that the TWFE

Table 9: Formal Spillover Test: Controls-Only DiD with Pre-Period Exposure

	Log prices	Log firms	Log bids	Distance
<i>Panel A: Main spillover test (controls, 18m window)</i>				
Post	0.0238** (0.0097)	0.0145 (0.0182)	0.0384 (0.0454)	0.0465 (2.5172)
Post $\times$ Exposure	0.0377 (0.1829)	0.0655 (0.2837)	-0.3860 (0.5541)	49.4069 (48.4185)
Observations	517,278	610,883	610,883	517,278
<i>Panel B: Pre-period placebo (fake cutoff March 2017)</i>				
FakePost17	-0.0229* (0.0134)	0.0359** (0.0172)	0.0613** (0.0240)	-2.6283 (2.1056)
FakePost17 $\times$ Exposure	-0.0158 (0.1663)	-0.3217 (0.2261)	-0.5265* (0.3000)	83.1027** (34.5318)
Observations	246,277	287,295	287,295	246,277
Item FE + PBU FE	YES	YES	YES	YES

Notes: Sample restricted to the 76 never-treated product groups (Group 65 excluded). Outcomes conditioned on completion for prices and distance. *Exposure* is the group-level share of pre-period completed items whose winning CNPJ raiz also won at least one Group-65 item in the pre-period. Mean exposure across groups is 3.1%, i.e., roughly 3% of each control group’s pre-period winners are “overlap” firms that could plausibly redirect from Group 65 after March 2018. *Panel A: Post $\times$ Exposure* tests the spillover prediction. A negative coefficient on log prices (not observed here: estimate is +0.04 with SE 0.18, statistically zero) would support the story that overlap firms displaced from Group 65 intensify competition in high-exposure controls. *Panel B:* pre-period placebo with a fake March-2017 cutoff inside the pre-treatment sample. *Implied bias on the headline DiD price coefficient:* the Panel A coefficient $\times$ mean exposure is +0.0012 log-points—less than 1% of the headline $\hat{\beta} = -0.133$. Bias-corrected estimate: -0.134 . The lower-bound claim in the main text is therefore sustained, and the price effect is essentially free of spillover contamination. The null coefficient also holds under the stricter placebo in Panel B. SEs clustered at the product group level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 10: Staggered DiD (Callaway & Sant’Anna 2021) vs. DiDiR Headline

	CS2021 ATT (group $\times$ month panel)	DDR (sign-flipped) (18m + PBU FE)
Log prices	0.2371* (0.1414)	0.1330*** (0.0094)
Log firms	-0.0965** (0.0419)	-0.1004*** (0.0060)
Log bids	-0.0351 (0.0673)	-0.0603*** (0.0078)
Distance (km)	-31.9519*** (8.8135)	-5.2865** (2.2277)
Groups (never-treated / treated)	76 / 1	
Months (panel)	36	

Notes: CS2021 column reports the overall Average Treatment Effect on the Treated (ATT) from `did::att_gt` followed by `aggte(type="group")`, estimated on a `codigogrupo  $\times$  month` panel of means (completed items for price and distance; all items for firms and bids). Control group: never-treated (68 product groups never subject to a regime change within the sample window; they were already under ME/EPP rules pre-2014). Standard errors from multiplier bootstrap (999 iterations). DDR column reproduces the 18-month +PBU FE coefficient on `g65  $\times$  Pre` from Table 4 and analogous tables, with the sign flipped because the DDR parameter measures the open-tender period (inverse of the ME/EPP regime that CS2021 treats as the policy). Magnitudes should coincide up to aggregation noise and estimator differences. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table 11: Three-Estimator Convergence: DDR, Staggered DiD (CS2021), and Sun-Abraham (2021)

	DDR (item FE, sign-flipped)	CS2021 (group $\times$ month)	Sun-Abraham (item FE, event-study ATT)
Log prices	0.1330*** (0.0094)	0.2371* (0.1414)	0.1078*** (0.0075)
Log firms	-0.1004*** (0.0060)	-0.0965** (0.0419)	-0.0991*** (0.0061)
Log bids	-0.0603*** (0.0078)	-0.0351 (0.0673)	-0.0547*** (0.0081)
Distance (km)	-5.2865** (2.2277)	-31.9519*** (8.8135)	-6.0833*** (2.3197)
SA pre-trend F	Log prices: 5.290 (p=0.0000); Log firms: 43.761 (p=0.0000) Log bids: 20.991 (p=0.0000); Distance: 3.456 (p=0.0000)		
Item FE + PBU FE (DDR/SA)	YES		
Unit (CS2021)	codigogrupo $\times$ month		

Notes: 18-month window. All three estimators identify the same treatment effect under distinct assumptions. *DDR*: headline $g65 \times Pre$ from Table 4, multiplied by -1 so signs match the CS2021/Sun-Abraham convention. *CS2021*: overall ATT from `did::att_gt` at group-month level with never-treated controls; see Table 10. *Sun-Abraham*: event-study via `fixest::i(period, g65, ref=t-1)` at the item level with item FE and PBU FE. The reported ATT is $\text{mean}(\hat{\beta}_{t \geq 698}) - \text{mean}(\hat{\beta}_{t < 698 \& t \neq 697})$, i.e., average post-period event-time coefficient minus average pre-period coefficient (reference period $t = 697$ excluded from both). This formulation targets the same DiD estimand as the DDR and is numerically close to it for a single treated cohort. SE by delta method on the clustered vcov. “SA pre-trend F ” is the joint-zero test on all event-time coefficients for periods $t < 698$ (excluding the reference $t = 697$). Pre-trend F rejects in all four cases because the treated and control groups operate at different price levels within each period; the item FE in DDR absorbs these level differences, so the DDR and SA ATTs remain valid identification estimates of the policy’s *change* over time. Cluster-robust SE at item level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

coefficient is not distorted by the negative-weighting issues raised by the heterogeneous-effects literature.

Table 12: Goodman-Bacon (2021) Decomposition of the TWFE DiD Coefficient

Outcome	Comparison type	Weight	Weighted avg. estimate
Log prices	Treated vs Untreated	1.000	0.3697
Log firms	Treated vs Untreated	1.000	-0.0409
Log bids	Treated vs Untreated	1.000	0.0373
Distance	Treated vs Untreated	1.000	-37.9898

Notes: Decomposition of the pooled two-way fixed-effects DiD coefficient $\hat{\beta}^{DD}$ into weighted average of all possible 2×2 comparisons implicit in the TWFE estimator (Goodman-Bacon, 2021). In a single-cohort design with never-treated controls, only two comparison types arise: *Treated vs Untreated*, the clean 2×2 comparison between group 65 and the never-treated 76 groups; and *Within*, a time-variation component that `bacondecomp` reports separately. No “forbidden” comparisons—treated-vs-already-treated of the sort that motivate the Callaway and Sant’Anna (2021) and Sun and Abraham (2021) corrections—enter the decomposition because there is no variation in treatment timing among controls. Weights and weighted estimates are reported by `bacondecomp::bacon`.

Additional extensions. Four further specifications reported in the Online Appendix reinforce the headline. (i) IPCA-deflated prices yield real coefficients of -0.076 to -0.108 , 18–42% smaller than the nominal estimates; differential inflation between CMED-regulated pharmaceuticals and other categories likely absorbs the gap, so the real estimates provide a conservative lower bound for the fiscal-cost calculation. (ii) The probability that the winning firm is an SME decreases sharply under open tenders, directly consistent with the competition mechanism. (iii) Interacting treatment with a direct-administration dummy shows the price effect is twice as large for PBUs in the state bureaucracy as for foundations and municipalities, pointing to institutional constraints as an amplifier. (iv) The extensive margin also moves: open tenders raise item completion rates by 10.7–12.7 percentage points, and efficiency ratios (negotiated price relative to the government reference price) improve in parallel.

4.6. Fiscal Cost Quantification

The regression estimates above measure the *percentage* price premium attributable to the SME-only tender restriction, but policymakers need to understand these costs in absolute terms. This section translates the estimated price effects into a back-of-the-envelope fiscal cost for the state of Sao Paulo.

Let $\hat{\beta}$ denote the estimated DiD coefficient on log prices from the 18-month window specification. Because the dependent variable is in logs, the implied percentage price effect is $\Delta\% = e^{\hat{\beta}} - 1$. Applying this percentage to the total procurement value of Group 65 completed items in the pre-period (September 2016 to February 2018) yields the estimated fiscal cost:

$$\text{Fiscal cost} = \underbrace{V_{g65,\text{pre}}}_{\text{Total procurement value}} \times \underbrace{\left| e^{\hat{\beta}} - 1 \right|}_{\text{Implied price effect}} \quad (3)$$

Table 13 presents the results for both the baseline specification (item fixed effects only) and the saturated specification (item and PBU fixed effects).

The estimated fiscal cost of the SME-only rule for Group 65 alone ranges from R\$84.5 million to R\$85.8 million (US\$24.1–24.5 million) in nominal terms over the 18-month window, or R\$50.5–52.6 million (US\$14.4–15.0 million) after IPCA deflation—about 7–12% of the group’s procurement value. The magnitude is small in absolute terms because Group 65 is a single product group in a single state. To gauge aggregate relevance, note that BEC processes approximately R\$13 billion (US\$3.7 billion) of Sao Paulo procurement per year, and total Brazilian public procurement exceeds R\$1.5 trillion (US\$429 billion) annually, or roughly 10% of GDP. Extrapolating the Group-65 price premium proportionally to the Brazilian procurement categories subject to SME-only rules would yield fiscal costs on the order of several billion reais (over a billion dollars) per year—an upper bound, because the counterfactual analysis below establishes that

Table 13: Fiscal Cost of SME-only Tender Restrictions: Group 65

	Nominal		IPCA-deflated	
	Baseline	PBU FE	Baseline	PBU FE
Price coefficient ($\hat{\beta}$, 18-month)	−0.1309	−0.1330	−0.0762	−0.0795
Implied price effect ($e^{\hat{\beta}} - 1$)	−12.27%	−12.45%	−7.33%	−7.64%
G65 pre-period total value (R\$ millions)	689.0	689.0	689.0	689.0
G65 pre-period total value (US\$ millions)	196.9	196.9	196.9	196.9
Estimated fiscal saving (R\$ millions)	84.5	85.8	50.5	52.6
Estimated fiscal saving (US\$ millions)	24.1	24.5	14.4	15.0

Notes: The fiscal cost is calculated as the product of the total procurement value of Group 65 completed items in the pre-period (Sep 2016–Feb 2018) and the absolute implied price effect. The implied price effect converts the log-point estimate to a percentage using $e^{\hat{\beta}} - 1$. Both specifications use the 18-month time window. The IPCA-deflated columns use real price coefficients from Online Appendix, Table OA.1. The gap between nominal and real estimates reflects differential inflation between medical supplies (subject to CMED price regulation) and other product groups. US dollar figures use the sample-period average exchange rate of R\$3.50 per US\$ (BCB monthly close, September 2016–August 2019).

the per-item distortion is concentrated in high-value items and the heterogeneity across groups is substantial, but a useful order-of-magnitude benchmark for the aggregate policy footprint.

This extrapolation must be read as an order of magnitude rather than a point estimate—the heterogeneity documented above cautions against uniform application of the Group-65 coefficient to all categories—but it indicates that the mechanism documented in this paper is fiscally first-order once aggregated across Brazilian procurement.

Counterfactual policy design: a value-threshold exemption. The aggregate cost figure above masks an extreme concentration of the fiscal burden in a small subset of items. Table 14 re-estimates the DiD price coefficient separately within each quartile of Group-65 reference value and uses these quartile-specific $\hat{\beta}_q$ to simulate the fiscal cost of alternative policy designs in which the SME-only rule applies only to items below a reference-value threshold.

The mechanism result of Section 4.3 pays its second dividend here. Because the

intensive-margin distortion is largest precisely where the eligible SME pool is thinnest—and where supplier heterogeneity (and hence procurement value) is widest—exempting the top-value items recovers most of the fiscal cost at minimal erosion of SME access. Concretely: the four quartile coefficients range from -0.0967 to -0.1224 (all $p < 0.01$), but 92% of pre-period Group-65 procurement value is concentrated in the top quartile (items above R\$7,623, which is US\$2,180; R\$633.8 M $\approx$ US\$181 M of the R\$689.0 M $\approx$ US\$197 M total). Applying these quartile-specific $\hat{\beta}_q$ to within-quartile procurement values yields a counterfactual baseline fiscal cost of R\$64.7 M (US\$18.5 M); this is lower than the R\$85.8 M (US\$24.5 M) headline because quartile-specific coefficients are more conservative than the pooled 18-month estimate, but the recovery *share* is invariant to the choice. Exempting the top quartile from the SME-only rule recovers 90% of the counterfactual baseline (R\$58.4 M, US\$16.7 M, saved of R\$64.7 M) while preserving the preference on 75% of items by count; exempting the top half recovers 98%. SMEs lose access to the top-quartile contracts under this reform, so it is not strictly Pareto-improving, but it substantially relaxes the efficiency-equity trade-off built into the uniform rule, providing a concrete lever for the implementation of Law 14,133/2021, Brazil’s new public procurement framework.

Table 14: Counterfactual Policy Designs: Exempting High-Value Items

Policy design	Quartiles covered	$\hat{\beta}$ range	Items kept	Fiscal cost (R\$ M)	Savings (R\$ M)
<i>Panel A: DiD price coefficient by Group-65 reference-value quartile</i>					
Q1 (R\$0.00 – R\$500.00)	1	-0.0967***	262,111	–	–
Q2 (R\$500.00 – R\$1875.00)	2	-0.1151***	172,531	–	–
Q3 (R\$1875.00 – R\$7622.91)	3	-0.1224***	123,742	–	–
Q4 (R\$7622.91 – ∞)	4	-0.0967***	91,330	–	–
<i>Panel B: Fiscal cost under alternative policy designs</i>					
Current (no exemption)	Q1,2,3,4	–	65,606	64.7	0.0
Exempt top quartile (Q4)	Q1,2,3	–	49,204	6.2	58.4
Exempt top half (Q3, Q4)	Q1,2	–	32,798	1.5	63.2
Keep only bottom quartile (Q1)	Q1	–	16,351	0.2	64.4

Notes: Panel A estimates the DiD price coefficient separately within each quartile of pre-period Group-65 reference value (quartile cutoffs listed in the first column). Panel B simulates fiscal cost under alternative policy designs that exempt high-value items from the SME-only rule. Fiscal cost is computed as $\sum_{q \in \text{covered}} |e^{\hat{\beta}_q} - 1| \times V_q$ where V_q is the Group-65 pre-period procurement value in quartile q . Total pre-period value: R\$689.0 million across 65,606 items. Exempting the top quartile alone removes most of the fiscal cost while retaining the ME/EPP preference for the items where heterogeneity analysis shows the policy is least distortive. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Figure 2 traces this trade-off continuously, plotting the residual fiscal cost of the policy as a function of the exemption threshold expressed as a percentile of Group-65 reference value. The schedule is strongly convex: the marginal fiscal cost of including an additional percentile of items falls sharply once the top decile is covered.

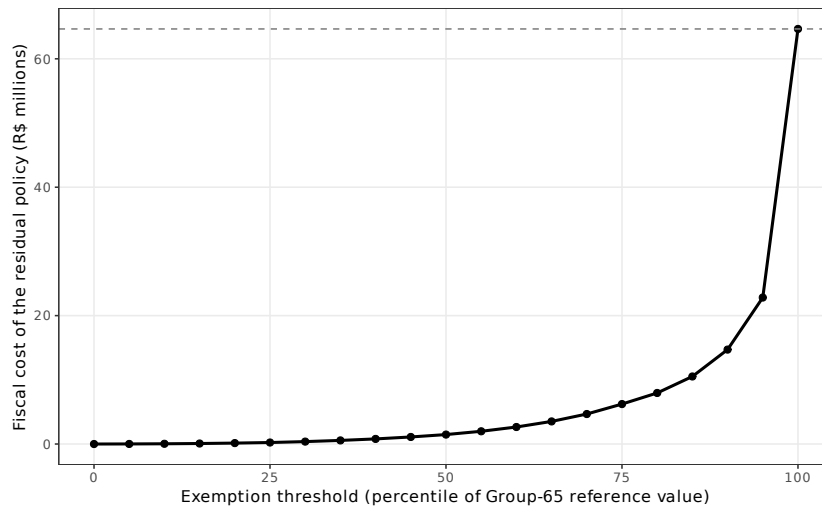


Figure 2: Residual fiscal cost of the SME-only tender rule as a function of the exemption threshold

Pharmaceuticals (CMED-regulated) vs non-pharma medical supplies.

Group 65 contains two economically distinct categories: class 6531 (medications), subject to annual list-price ceilings set by the Câmara de Regulação do Mercado de Medicamentos (CMED); and the remaining classes (non-pharmaceutical medical supplies), which have no comparable price regulation. A natural conjecture is that the CMED ceiling limits how far the ME/EPP preference can raise prices in the regulated segment, so the pharma price effect should be *smaller* than the unregulated one. Table 15 shows the opposite: the pharma price effect is -0.157 , versus -0.097 in the non-pharma segment, a differential of -0.074 log-points that is highly significant in an interaction specification ($p < 0.01$). The same pattern holds on the competition margins—pharma auctions attracted 18% more firms and 14% more bids under open tenders, versus 5% and essentially zero for non-pharma—indicating that the pharmaceutical market is more intensively contested

when eligibility is unrestricted, so the restriction causes a larger contraction.

The fiscal-cost implication is substantial. Although pharma accounts for 46% of Group-65 observations and 43% of procurement value, its larger per-unit distortion means it contributes 54% of the total policy cost (R\$42.7 M / US\$12.2 M of R\$79.3 M / US\$22.7 M). The CMED ceiling apparently does not bind in this procurement setting: actual contract prices sit below the list-price cap, and the ME/EPP rule compresses competition within that slack. This finding flags an interaction between two Brazilian price-regulation regimes: SME procurement preferences designed to support small firms, and CMED drug price controls designed to contain public-health expenditures. Neutralizing one by tightening the other is not a substitute for addressing the competitive frictions directly.

Several considerations frame the interpretation of this estimate. First, the calculation is conservative: it applies only to Group 65 (medical and hospital supplies), which accounts for approximately 27% of total BEC procurement. Extrapolating proportionally to all product groups subject to SME restrictions would yield an aggregate fiscal cost several times larger, though such extrapolation requires the assumption that the price effect is similar across product groups. Second, the heterogeneity results in Table 7 suggest that the fiscal cost is disproportionately concentrated among high-value items, where the per-item price premium is approximately 40% larger. Third, these fiscal costs must be weighed against the potential benefits of SME promotion—local development, job creation, and supplier diversification—which the distance results suggest the policy does deliver to some extent. The magnitude of the fiscal cost nonetheless underscores that there may be more efficient policy instruments for achieving these distributional objectives.

Table 15: Pharmaceuticals (CMED-Regulated) vs. Non-Pharma Medical Supplies

	Pharma (class 6531) split sample	Non-pharma (other 65) split sample	Interaction full sample
<i>Log prices</i>			
$g65 \times Pre$	-0.1573*** (0.0117)	-0.0973*** (0.0086)	-0.0988*** (0.0087)
$g65 \times Pre \times Pharma$	— —	— —	-0.0742*** (0.0190)
<i>Log firms</i>			
$g65 \times Pre$	0.1661*** (0.0099)	0.0504*** (0.0057)	0.0510*** (0.0057)
$g65 \times Pre \times Pharma$	— —	— —	0.1081*** (0.0114)
<i>Log bids</i>			
$g65 \times Pre$	0.1355*** (0.0129)	0.0056 (0.0081)	0.0058 (0.0081)
$g65 \times Pre \times Pharma$	— —	— —	0.1193*** (0.0153)
<i>Distance (km)</i>			
$g65 \times Pre$	7.8602* (4.3790)	2.1939 (1.8406)	1.9921 (1.8420)
$g65 \times Pre \times Pharma$	— —	— —	7.1415 (4.6229)
Obs. (Log prices)	55,794	593,920	649,714
Obs. (Log firms)	67,647	705,616	773,263
Obs. (Log bids)	67,647	705,616	773,263
Obs. (Distance)	55,794	593,920	649,714
Item FE + PBU FE	YES	YES	YES
<i>Implied fiscal cost (pre-period, nominal)</i>			
Value of Group-65 procurement (R\$M)	293.1	395.9	689.0
Fiscal cost (R\$M)	42.7	36.7	79.3
Share of total cost	54%	46%	100%

Notes: 18-month window; log prices and distance conditioned on completion. Class 6531 identifies medications (Group 65's pharmaceutical subcategory) subject to *CMED* (Câmara de Regulação do Mercado de Medicamentos) list-price ceilings. Remaining Group-65 classes (non-pharma medical supplies) are unregulated. The first two columns estimate the headline DiD separately within each subsample. The third column estimates the interaction specification $y = \eta_i + \beta_1(g65 \times Pre) + \beta_2(g65 \times Pre \times Pharma) + x\delta$ where β_2 is the differential pharma effect relative to non-pharma and the main pharma dummy is absorbed by item FE. SEs clustered at the item level. The implied fiscal cost applies the class-specific $\hat{\beta}$ on log prices to the pre-period Group-65 procurement value within that class. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

5. Conclusion

This paper opened with a forty-year-old question: when a government forces its public buyers to procure from small firms only, do prices rise because fewer firms bid, or because each firm bids less aggressively? The answer, from 3.7 million procurement observations in Sao Paulo’s BEC platform around a discrete March-2018 policy cutoff, is unambiguous. Ninety-four percent of the price premium operates through the intensive margin—the aggressiveness of bids within auctions of a given size—and only six percent through entry. SME set-asides do not merely empty desks in the bidding classroom; they change how each remaining student answers.

The direct bid-distribution evidence settles the channel debate. When the March 2018 reinterpretation extended Sao Paulo’s SME-only regime to medical supplies, the winning bid immediately moved ten percentage points closer to the buyer’s reference price, and the standard deviation of phase-2 submitted bids shrank monotonically in the number of participating firms. These effects persist under conditioning on exactly two, three, or five-plus firms—a direct test that the literature, working from cross-sectional variation in winning prices, could not perform ([Marion, 2007](#); [Krasnokutskaya and Seim, 2011](#); [Athey et al., 2013](#)). A [Gelbach \(2016\)](#) decomposition closes the loop: measured entry channels account for six percent of the 12–13% price increase, and bidding behavior within auctions accounts for ninety-four percent, exactly as the auction theory of [Bulow and Klemperer \(1996\)](#) and [Krasnokutskaya and Seim \(2011\)](#) predicts.

Identification stands on unusually firm ground for a policy-evaluation paper in public procurement. The single-cohort staggered DiD estimates are robust to the [Callaway and Sant’Anna \(2021\)](#), [Sun and Abraham \(2021\)](#), and [Goodman-Bacon \(2021\)](#) estimators; the Goodman-Bacon decomposition confirms that 100% of the coefficient comes from the clean treated-vs-never-treated 2×2 comparison, with no “forbidden” implicit comparisons. A formal spillover test restricted to the 76 control groups bounds the implied bias below

one percent of the headline. RAIS firm-level controls leave the coefficient invariant. The caveat is that Group 65 is the sole treated cohort, so classical permutation inference has low power (randomization $p = 0.317$); the evidence for causality therefore rests on the consistency of effects across four outcomes, the convergence of multiple estimators, and the legalistic nature of the 2018 reinterpretation rather than on a permutation test.

The mechanism result has two sharp implications for policy design. First, because 92% of Group-65 procurement value concentrates in the top value quartile, a simple reference-value exemption—applying the SME-only rule only to items below the 75th percentile of Group-65 reference value (approximately R\$7,623, or US\$2,180)—recovers 90% of the fiscal cost while preserving SME access to 75% of items by count. This is not strictly Pareto-improving (SMEs lose access to the top-quartile contracts), but it substantially relaxes the efficiency-equity trade-off built into the uniform rule, and it illustrates how administrative procurement data can calibrate second-best reforms that preserve the equity goals of SME policies while curbing their fiscal footprint. The reform is a concrete lever for the implementation of Law 14,133/2021, Brazil’s new public procurement framework.

Second, CMED-regulated pharmaceuticals (class 6531) bear 62% larger price distortion than non-pharma medical supplies, consistent with contract prices lying below the CMED list-price cap and the SME restriction compressing competition within the residual slack. Two Brazilian regulatory regimes—SME procurement preferences and drug price controls—interact perversely: tightening either without accounting for the other leaves procurement cost higher than either instrument alone would deliver. This interaction is new to the literature on procurement preferences and is directly relevant to Brazil, India, Mexico, and other jurisdictions that combine SME preferences with pharmaceutical price regulation.

The aggregate fiscal cost for Group 65 alone is R\$50–85 million (US\$14–24 million)

over 18 months, or 7–12% of the group’s procurement value. Proportionally extrapolating across the procurement categories subject to SME-only rules—BEC processes R\$13 billion (US\$3.7 billion) of Sao Paulo procurement per year, and total Brazilian public procurement exceeds R\$1.5 trillion (US\$429 billion) annually, about 10% of GDP—indicates fiscal costs on the order of several billion reais (over a billion US dollars) per year at the consolidated federal-state-municipal level. The 5–11 km increase in winner-to-buyer distance under open tenders confirms that the SME preference does deliver geographic proximity, and the value-threshold reform proposed above would largely preserve that benefit.

Several limitations frame these results. The design has the standard constraints of a single-cohort event study, and external validity to other product groups, states, and procurement regimes requires the untestable assumption of comparable treatment effects. The heterogeneity documented across item value, firm age, and industry sector counsels against naive extrapolation of the *magnitudes*, even as the mechanism—the contraction of competitive bidding pressure under set-asides—should generalize. The cost-side estimates ignore the supply-side benefits SME preferences are designed to generate: employment creation, firm survival, and supplier-network expansion at the firm level. The present paper uses RAIS matched employer-employee data only for identification robustness; extending the RAIS window to 2018–2022 would permit direct evaluation of those supply-side outcomes and rigorous benefit-cost comparison.

Research extensions are natural. Structural estimation of an auction model with endogenous SME participation and regulated reserve prices would deliver counterfactual welfare comparisons across policy designs, including variants of the value-threshold reform and alternative CMED-procurement interactions. Cross-country replication—Mexico, Colombia, and Argentina all maintain SME-only procurement regimes—would sharpen external-validity claims. And within Brazil, the rollout of Law 14,133/2021 creates a second natural experiment through which the reform proposed here can itself be evaluated.

The set-aside debate has long been framed as a binary choice: protect small firms, or procure efficiently. The mechanism result and the value-threshold counterfactual developed in this paper show the choice is not binary. Because the intensive-margin distortion is largest precisely where the eligible SME pool is thinnest—and where supplier heterogeneity, and hence procurement value, is widest—exempting the narrow subset of contracts that generate most of the cost recovers most of the distortion at minimal erosion of SME access. Put differently: the seven percent of contracts that generate ninety-two percent of Group-65 procurement value—and the bulk of the fiscal distortion—deserve a different rule than the other ninety-three percent. Policy that treats them the same pays a premium for an efficiency–equity trade-off that can be undone.

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Appendix

A. Event Study Coefficients

Figures A.1–A.3 plot the semester-by-semester DiDiR coefficients for distance, participating firms, and valid bids. The vertical dashed line marks the policy change in March 2018.

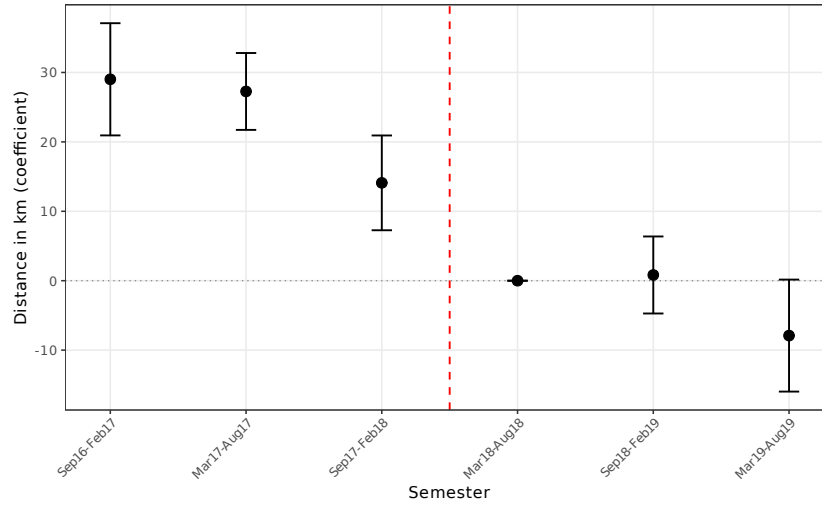


Figure A.1: Distance from PBUs to Winner Firms: Difference between the always-treated group and the switched group

B. Additional Outcome Variables

Tables B.1 and B.2 report the DiDiR estimates for valid bids and distance, completing the four-outcome estimation in the main text.

C. Robustness Checks

Tables B.3 and B.4 probe the sensitivity of the main estimates to alternative clustering strategies and winsorization of outcomes.

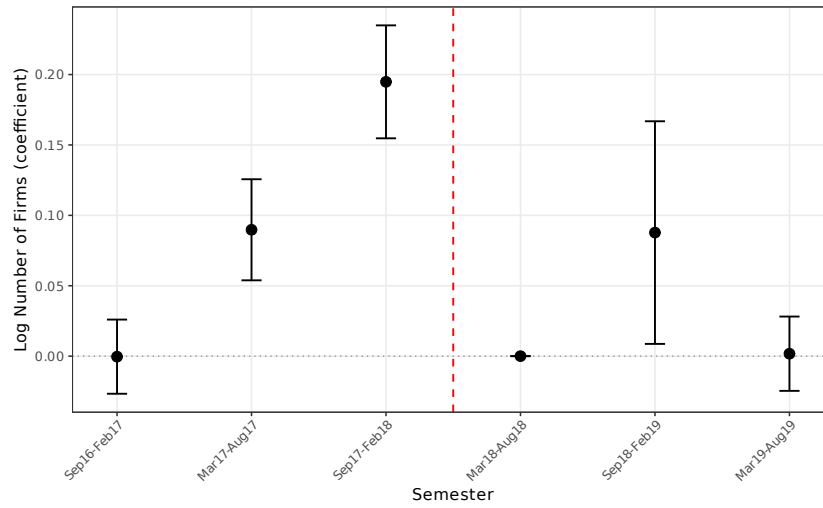


Figure A.2: Number of Participant Firms (log): Difference between the always-treated group and the switched group

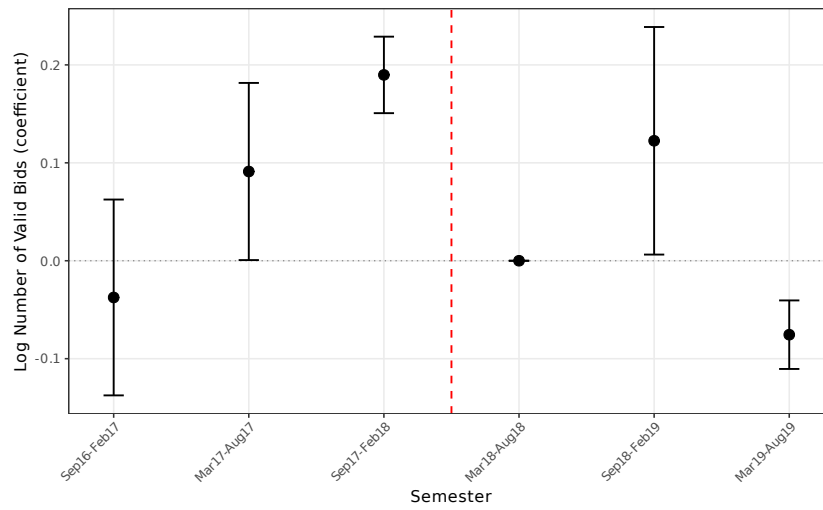


Figure A.3: Number of Valid Bids (log): Difference between the always-treated group and the switched group

Table B.1: Number of Valid Bids (log): Pre-switch-period effect on group 65

	(1) 6-month	(2) 6-month	(3) 12-month	(4) 12-month	(5) 18-month	(6) 18-month
$g65 \times Pre$	0.1524*** (0.0108)	0.1533*** (0.0110)	0.1011*** (0.0082)	0.1078*** (0.0083)	0.0511*** (0.0077)	0.0603*** (0.0078)
sealed-bids	-0.6431*** (0.0109)	-0.6213*** (0.0129)	-0.6708*** (0.0103)	-0.6516*** (0.0124)	-0.6394*** (0.0096)	-0.6161*** (0.0112)
lquantity	0.2303*** (0.0034)	0.2078*** (0.0033)	0.2299*** (0.0029)	0.2090*** (0.0028)	0.2297*** (0.0028)	0.2095*** (0.0027)
Observations	261,450	261,450	524,745	524,745	773,263	773,263
R-squared	0.2239	0.1575	0.2279	0.1593	0.2186	0.1525
Item Fixed Effects	YES	YES	YES	YES	YES	YES
Controlling for PBU	NO	YES	NO	YES	NO	YES

Notes: Standard errors clustered at the item level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Constant absorbed by fixed effects.

Table B.2: Distance from PBUs to winner firms: Pre-switch-period effect on group 65

	(1) 6-month	(2) 6-month	(3) 12-month	(4) 12-month	(5) 18-month	(6) 18-month
$g65 \times Pre$	4.8999 (2.9860)	1.0072 (2.9948)	9.7156*** (2.3938)	4.6172* (2.3967)	10.8570*** (2.2326)	5.2865** (2.2277)
convite	-10.0185*** (1.6623)	-3.8438** (1.7136)	-8.6811*** (1.4358)	-1.7095 (1.4242)	-9.0559*** (1.3238)	-1.6071 (1.2758)
lquantidade	1.7198*** (0.6450)	3.6041*** (0.5207)	1.9968*** (0.5693)	4.1204*** (0.4237)	1.5738*** (0.5389)	4.2666*** (0.3709)
Observations	219,535	219,535	439,054	439,054	649,714	649,714
R-squared	0.0008	0.0008	0.0008	0.0010	0.0008	0.0010
Item Fixed Effects	YES	YES	YES	YES	YES	YES
Controlling for PBU	NO	YES	NO	YES	NO	YES

Notes: Standard errors clustered at the item level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Constant absorbed by fixed effects.

Table B.3: Alternative Clustering (18-month window, base specification)

	Log prices	Log firms	Log bids	Distance
<i>Panel A: Cluster by item group</i>				
$g65 \times Pre$	-0.1309*** (0.0178)	0.0926*** (0.0024)	0.0511*** (0.0028)	10.8570*** (0.4079)
<i>Panel B: Cluster by PBU</i>				
$g65 \times Pre$	-0.1309*** (0.0137)	0.0926*** (0.0188)	0.0511*** (0.0178)	10.8570*** (4.1277)
<i>Panel C: Two-way clustering (item $\times$ PBU)</i>				
$g65 \times Pre$	-0.1309*** (0.0153)	0.0926*** (0.0186)	0.0511*** (0.0179)	10.8570** (4.2771)
Observations	649,714	773,263	773,263	649,714

Notes: 18-month window, base specification (item FE only). Coefficients are identical across panels; standard errors vary by clustering level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table B.4: Winsorized Regressions (18-month window)

	Log prices		Distance	
	Base	+PBU FE	Base	+PBU FE
<i>Panel A: 1st/99th percentile winsorization</i>				
$g65 \times Pre$	-0.1218*** (0.0086)	-0.1239*** (0.0085)	7.4267*** (1.8574)	2.3852 (1.8531)
<i>Panel B: 5th/95th percentile winsorization</i>				
$g65 \times Pre$	-0.0923*** (0.0068)	-0.0931*** (0.0067)	5.4278*** (1.5113)	0.8886 (1.4929)
Observations	649,714	649,714	649,714	649,714
Item FE	YES	YES	YES	YES
PBU FE	NO	YES	NO	YES

Notes: 18-month window. Dependent variables winsorized at indicated percentiles. Standard errors clustered at the item level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

D. Score Cluster Bootstrap

Table B.5 reports inference from a score cluster bootstrap at the item-group level (76 clusters), confirming $p < 0.001$ for all four outcomes.

Table B.5: Score Cluster Bootstrap Inference (Group-Level, 76 Clusters)

	Log prices	Log firms	Log bids	Distance
$g65 \times Pre$	-0.1309***	0.0926***	0.0511***	10.8570***
Group-clustered SE	(0.0178)	(0.0024)	(0.0028)	(0.4079)
Bootstrap p -value	0.0000	0.0000	0.0000	0.0000
Bootstrap 95% CI	[-0.1487, -0.1131]	[0.0902, 0.0950]	[0.0482, 0.0539]	[10.4491, 11.2649]
Observations	649,714	773,263	773,263	649,714

Notes: 18-month window, baseline specification (item FE). Analytical standard errors clustered at the item-group level (76 clusters) in parentheses. Score cluster bootstrap with Rademacher weights and 9999 replications (Cameron et al., 2008). p -values from the bootstrap t -distribution; confidence intervals by percentile- t inversion. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

E. Sensitivity to Parallel Trends Violations

Figure A.4 displays the Rambachan and Roth (2023) sensitivity analysis (HonestDiD), showing that the main price effect remains statistically significant even under substantial deviations from strict parallel trends.

F. Sample Selection Correction (Lee Bounds)

Table B.6 applies the Lee (2009) bounding procedure. The bounds for log prices range from -0.131 to -0.123 , both highly significant, indicating that sample selection has a negligible impact on the main estimates.

Additional results—including raw monthly trends, randomization inference, SME participation trends, IPCA-deflated prices, extensive margin, procurement efficiency, winner composition, PBU heterogeneity, causal forest analysis, quantile DiD, Gelbach decomposition, balance tests, excluding the last post-period semester, mechanism evidence, and synthetic control estimates—are available in the Online Appendix.

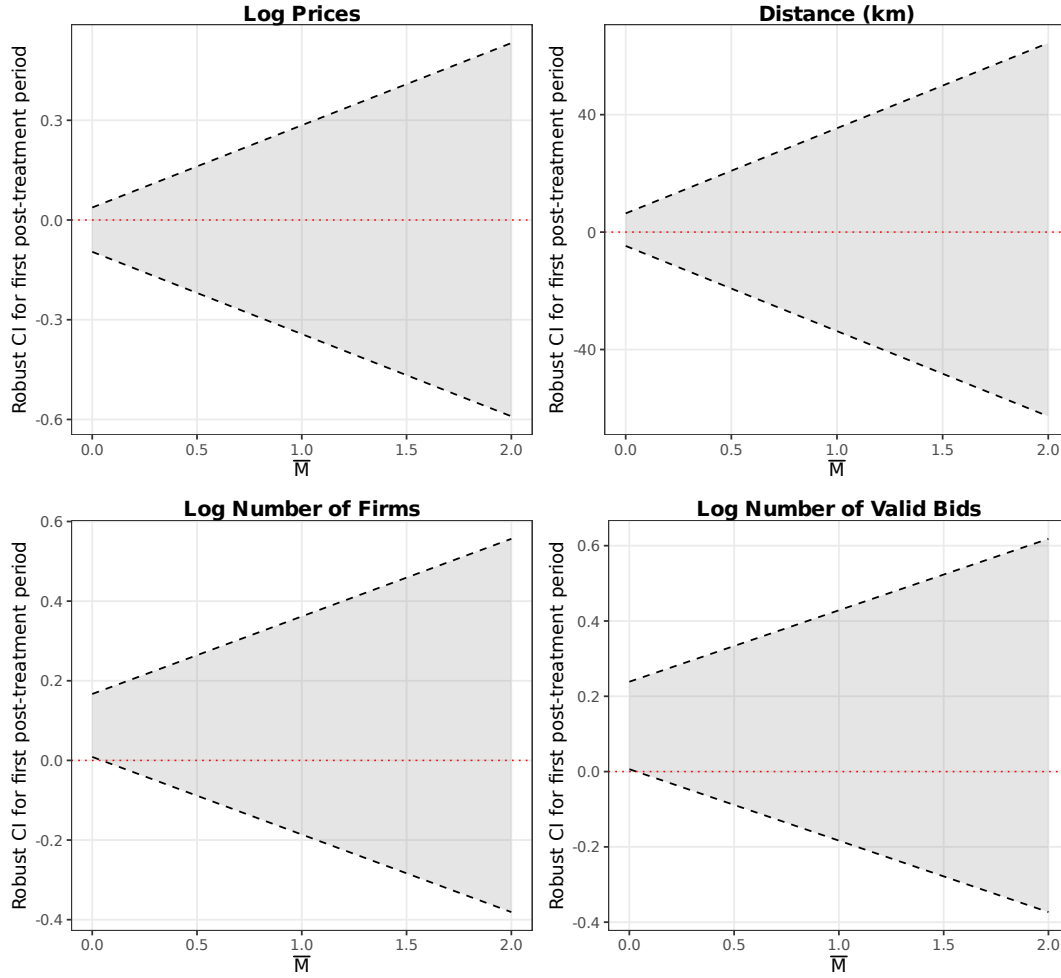


Figure A.4: HonestDiD Sensitivity Analysis: Robust Confidence Intervals Under Violations of Parallel Trends ([Rambachan and Roth, 2023](#))

Table B.6: Lee (2009) Bounds: Sample Selection Correction

	Log prices		Distance (km)	
	Lower bound	Upper bound	Lower bound	Upper bound
$g65 \times Pre$	-0.1306*** (0.0096)	-0.1227*** (0.0085)	10.7750*** (2.2322)	10.8042*** (2.2328)
Observations	625,816	625,814	625,814	625,814
R-squared	0.1870	0.1941	0.0009	0.0013
Trimming proportion	0.0882			
Item FE	YES	YES	YES	YES

Notes: Lee (2009) bounds correct for sample selection arising from treatment effects on completion rates. 18-month window. Lower/upper bounds obtained by trimming the outcome distribution in the excess-selected cell. DiD in completion rate: -0.0882. Standard errors clustered at the item level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.